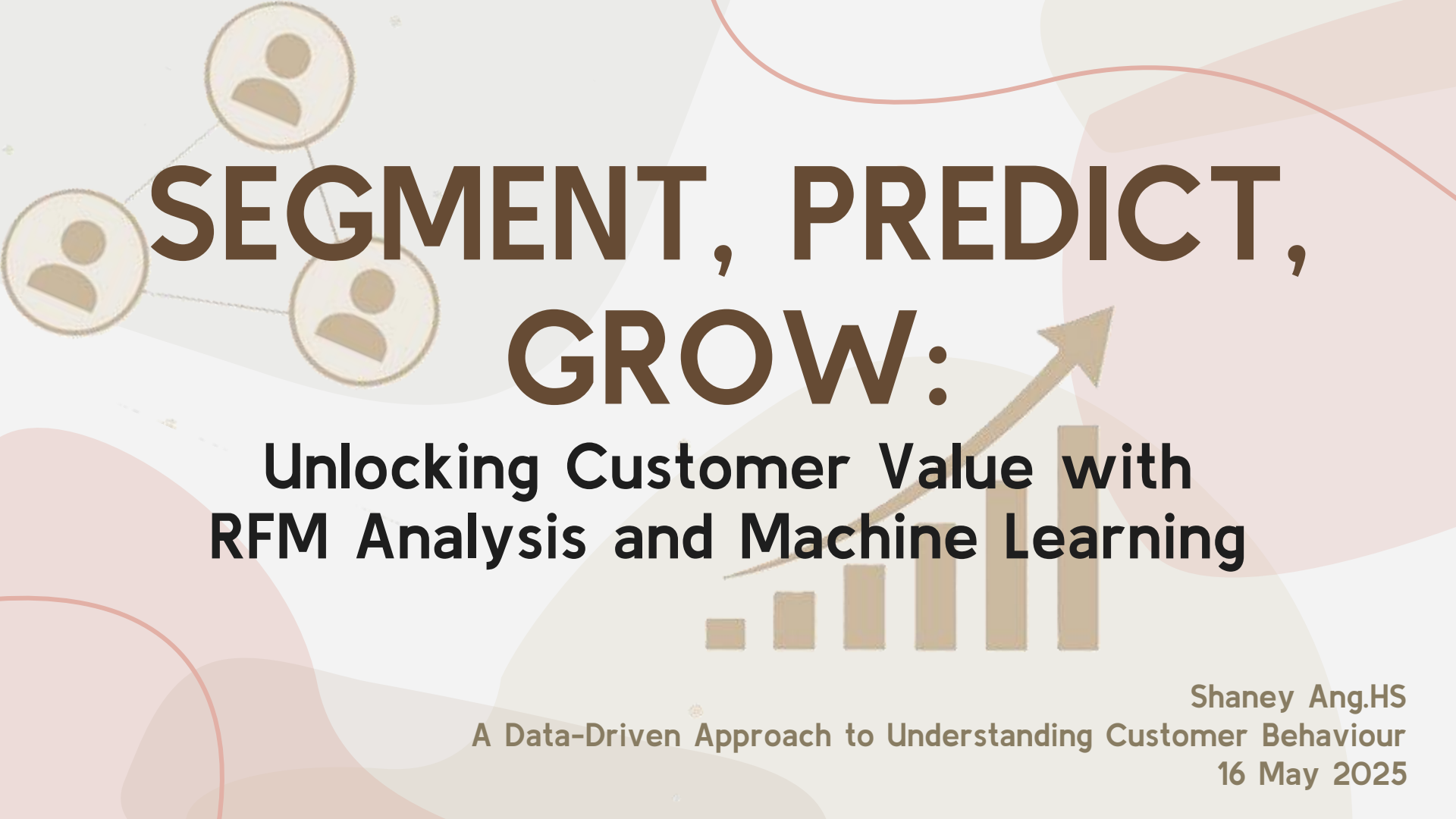




SEGMENT, PREDICT, GROW:

**Unlocking Customer Value with
RFM Analysis and Machine Learning**

Shaney Ang.HS
A Data-Driven Approach to Understanding Customer Behaviour
16 May 2025

AGENDA

Background

Establishing project context and problem statement



Scope

Defining objectives, stakeholder, and value proposition

Analytics

Preparing data and conducting exploratory analysis



Methodology

Applying RFM analysis, K-Means clustering, and regression modeling

Insights and Recommendations

Interpreting findings and proposing actionable solutions



CONTEXT

Dataset: UK Online Retail dataset

Source: <https://www.kaggle.com/datasets/vijayuv/onlineretail/data>

The dataset contains transactions from a UK-based online retail store selling all-occasion gifts.



*How many % of the
world's population
shop online?*

As of 2024, there are 2.71 billion online shoppers globally.¹

33%



Business Goal:

- Improve customer retention
- Increase marketing effectiveness
- Grow customer value

Business Challenge:

With thousands of customers and transactions, the business lacks a data-driven way to segment and prioritise customer relationships.

¹ <https://growthdevil.com/how-many-people-shop-online>

PROBLEM STATEMENT

How can we **segment customers** based on their purchasing behaviour to tailor marketing strategies, *and* **predict customer lifetime value (CLV)** to focus efforts on high-value customers?

SCOPE

Objectives:

This project analyses customer purchasing behavior using RFM metrics and machine learning to segment customers and predict their value.

The insights enable the marketing team to target high-value segments, improve retention, and drive revenue growth.

Key Questions:

1. Who are our most valuable customers, and how do they behave?
2. How can we segment customers for better marketing decisions?
3. How can we prioritise retention based on customer value?

STAKEHOLDER

– *Why it matters?*



Marketing Team

- 1. Turn Missed Revenue into Meaningful Relationships.**
- 2. No More Guesswork: Reach the Right Customer, Every Time.**
 - Enables smarter targeting, more effective campaigns.
 - Improve ROI by allocating budget to the most profitable segments.

ANALYTICS



DATA PREPARATION



Remove Nulls

Eliminate invoices with missing Customer IDs.



Remove Negatives

Discard cancelled invoices and related originals (negative values in quantity and price).



Engineer RFM

Construct recency, frequency, and monetary features.



Calculate CLV

Compute customer lifetime value using purchase proxy.



Feature Engineering

Create total quantity, average qty per order, tenure features.

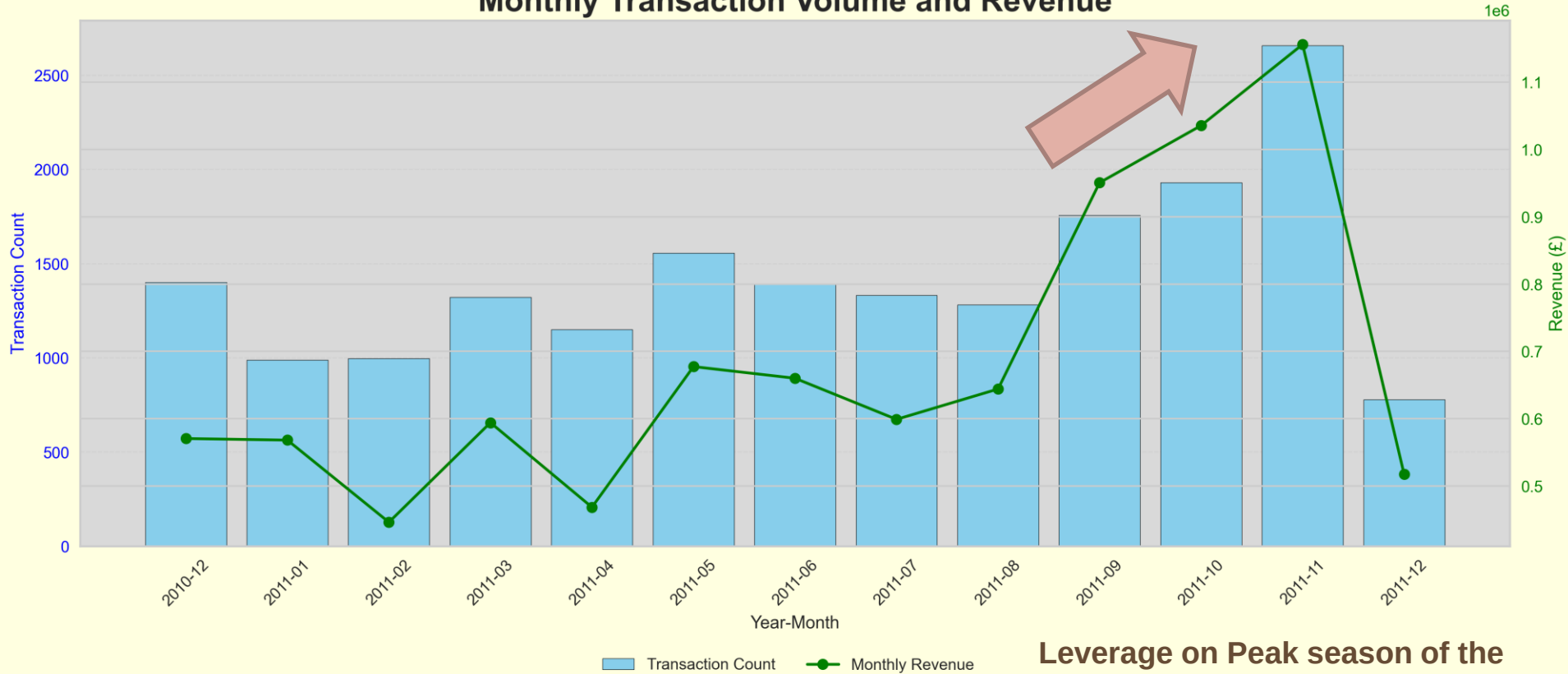
```
<class 'pandas.core.frame.DataFrame'>
Index: 392692 entries, 0 to 541908
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   InvoiceNo        392692 non-null object
1   StockCode       392692 non-null object
2   Description     392692 non-null object
3   Quantity        392692 non-null int64
4   InvoiceDate      392692 non-null datetime64[ns]
5   UnitPrice       392692 non-null float64
6   CustomerID      392692 non-null float64
7   Country         392692 non-null object
8   InvoiceYear      392692 non-null int32
9   YearMonth       392692 non-null object
10  TotalValue      392692 non-null float64
11  day_name        392692 non-null object
12  month_name      392692 non-null object
13  month           392692 non-null int32
14  day             392692 non-null int32
dtypes: datetime64[ns](1), float64(3), int32(3),
memory usage: 43.4+ MB
```

Cleaned Dataset

Engineered RFM Table:

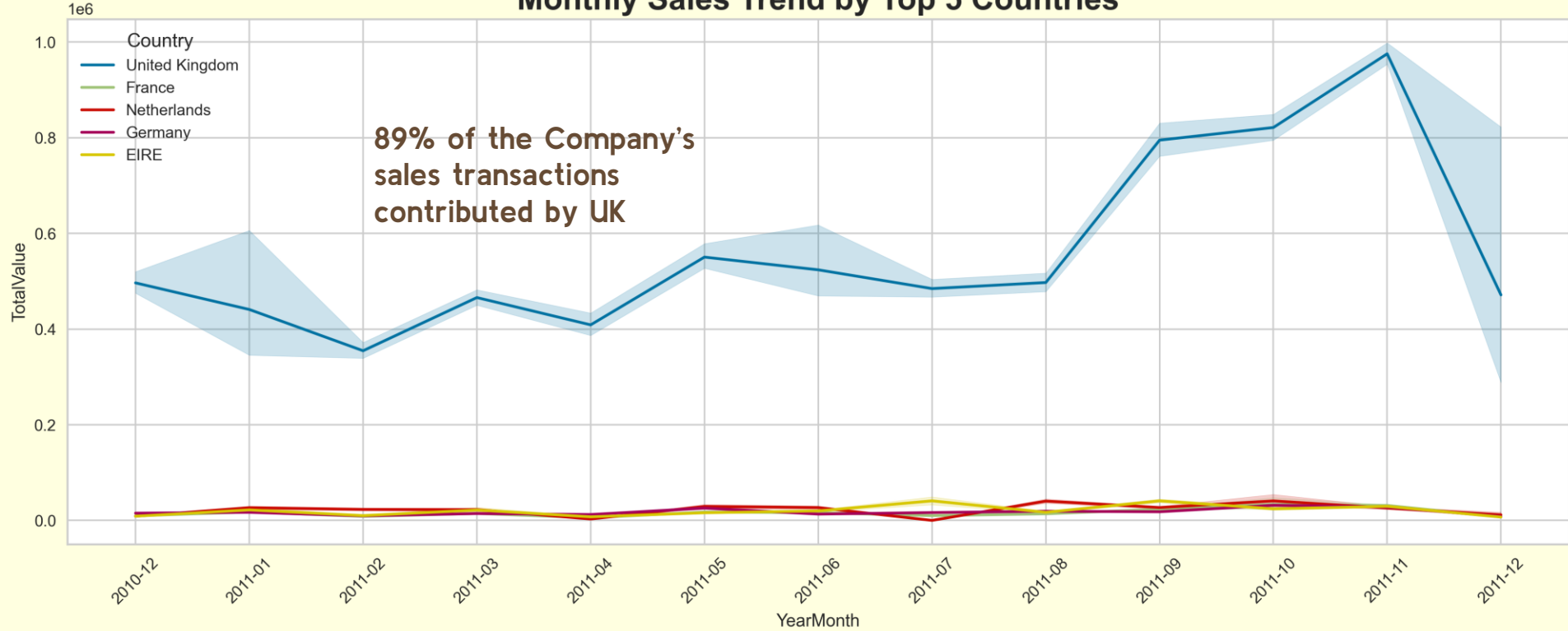
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4338 entries, 0 to 4337
Data columns (total 13 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   CustomerID      4338 non-null   float64
1   Recency         4338 non-null   int64
2   Frequency       4338 non-null   int64
3   Monetary        4338 non-null   float64
4   Country         4338 non-null   object
5   log_Recency     4338 non-null   float64
6   log_Frequency   4338 non-null   float64
7   log_Monetary    4338 non-null   float64
8   Cluster         4338 non-null   int32
9   AvgOrderValue   4338 non-null   float64
10  TotalQuantity   4338 non-null   int64
11  Tenure          4338 non-null   int64
12  AvgQuantityPerOrder 4338 non-null   float64
dtypes: float64(7), int32(1), int64(4), object(1)
memory usage: 423.8+ KB
```

Monthly Transaction Volume and Revenue

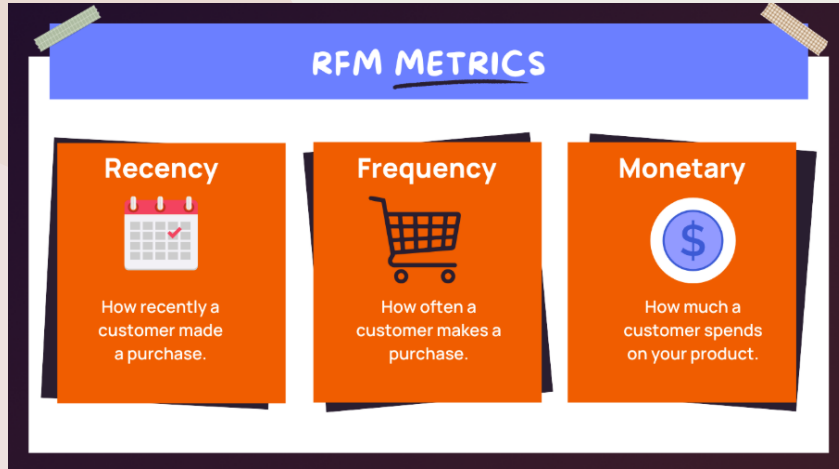


**Leverage on Peak season of the Company between Sep to Nov
→ e.g. Christmas Season**

Monthly Sales Trend by Top 5 Countries

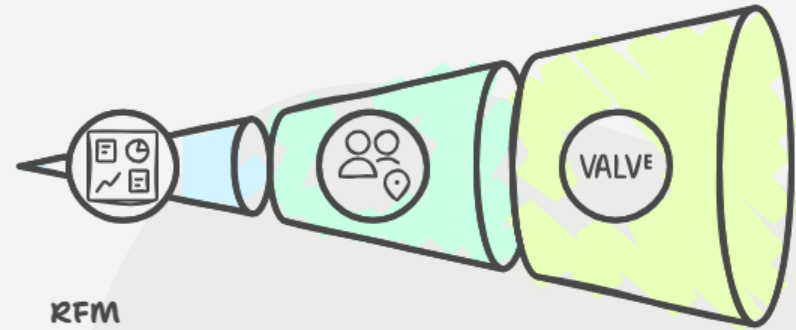


MODELING PROCESS WORKFLOW



CLV = Using total purchase value over the dataset as a *proxy CLV* for the predictive model.

Unlocking Customer Value



RFM Analysis

Analyzing customer behaviour using RFM metrics

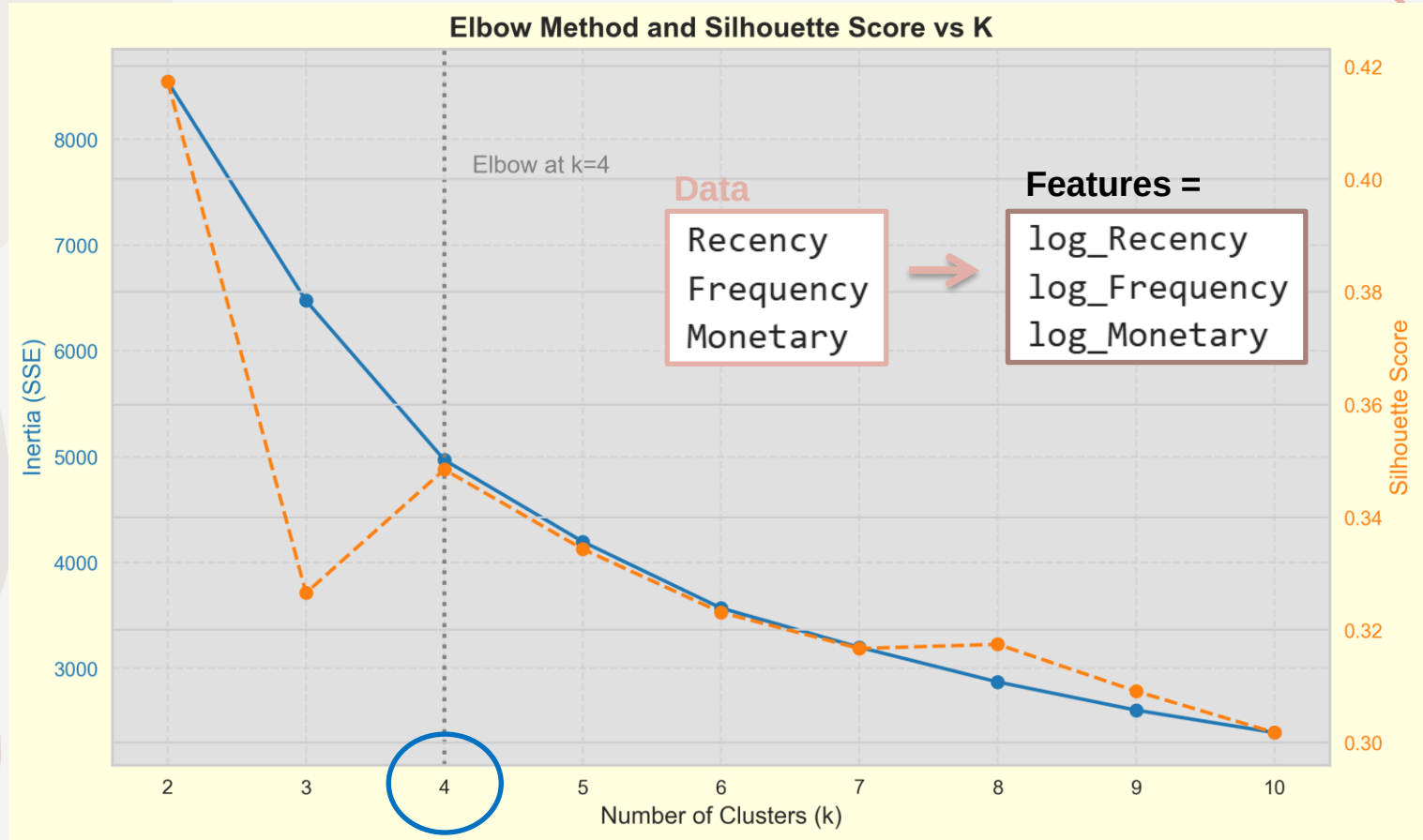
Customer Segmentation

Grouping customers using RFM values + K-Means Clustering

CLV Prediction

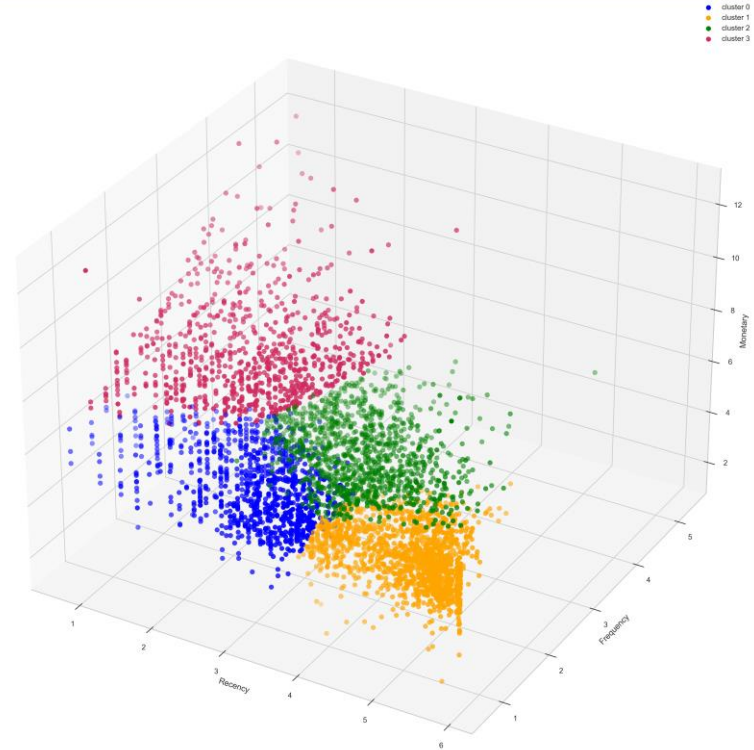
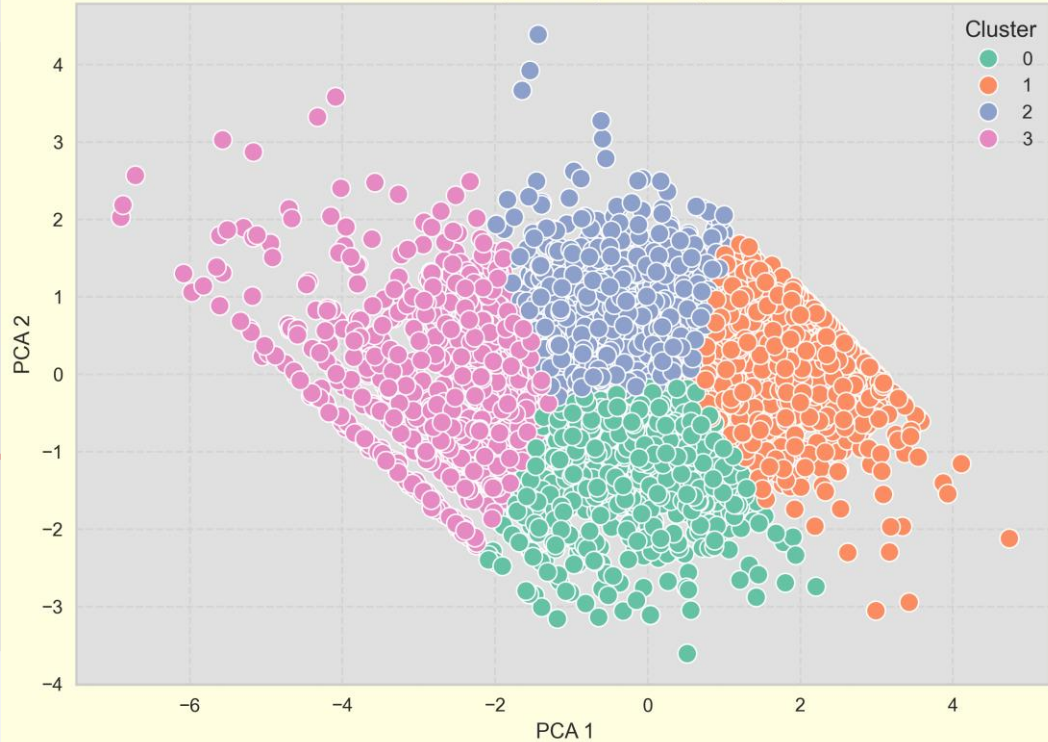
Predicting customer lifetime value using Supervised Machine Learning

K-MEANS CLUSTERING

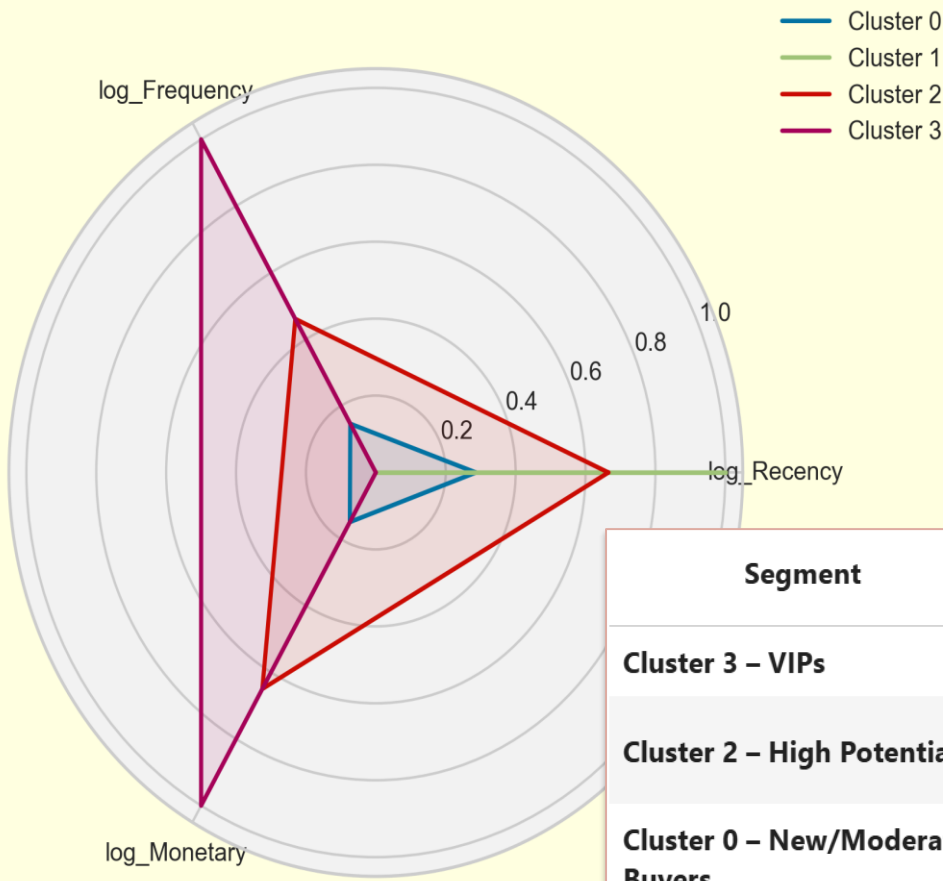


K-MEANS CLUSTERING

Customer Segments (PCA Projection)



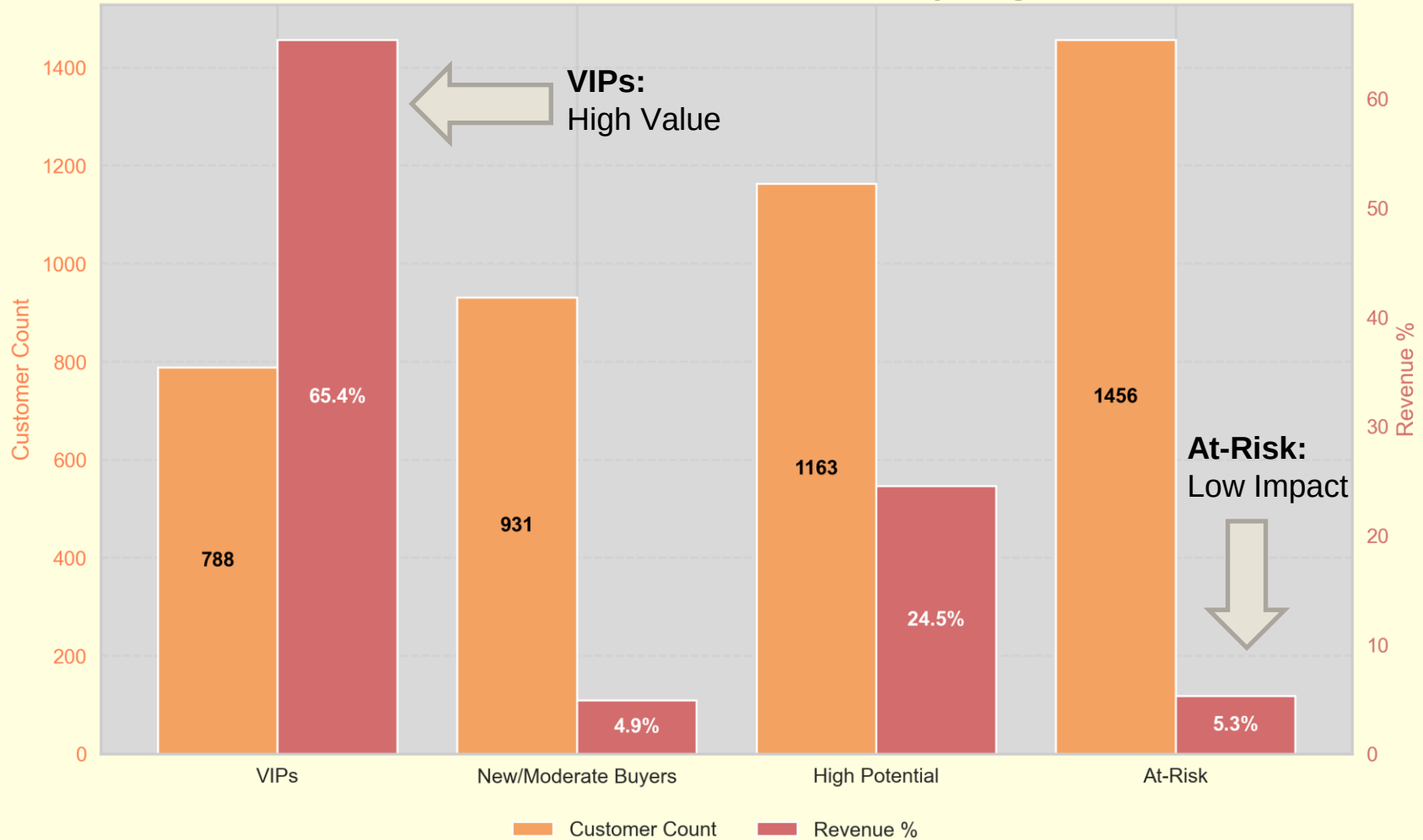
Log-RFM Profile by Cluster (Normalized)



INTERPRET CLUSTERS

Segment	% of Customers	Avg Order Value	Key Traits
Cluster 3 – VIPs	18%	\$594	Frequent, recent, high spend
Cluster 2 – High Potential	27%	\$456	Moderate recency, good spend but less frequent
Cluster 0 – New/Moderate Buyers	21%	\$222	Recent, low frequency, low-mid spend
Cluster 1 – At-Risk	34%	\$230	Inactive, low frequency, low spend

Customer Count and Revenue % by Segment



REGRESSION MODELING – CLV PREDICTION

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 4338 entries, 0 to 4337  
Data columns (total 13 columns):  
#   Column                Non-Null Count  Dtype    
---  ---                  
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2   Frequency              4338 non-null   int64    
3   Monetary               4338 non-null   float64  
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8   Cluster                4338 non-null   int32     
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11  Tenure                 4338 non-null   int64     
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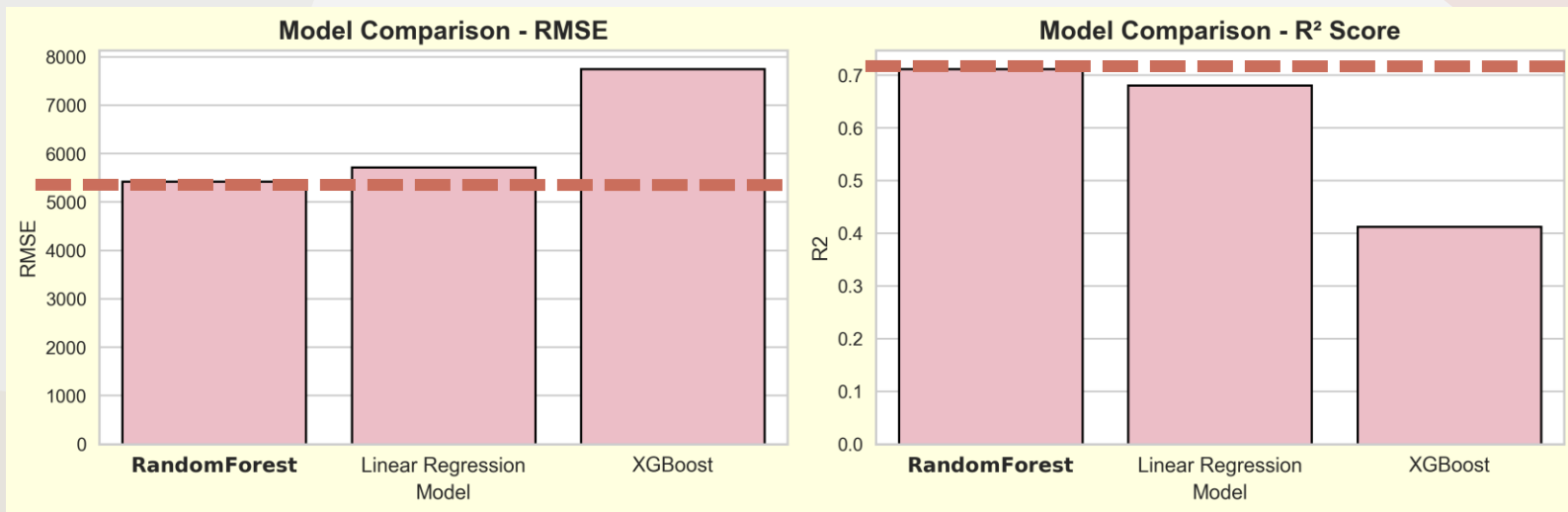
Output Y
(predict CLV proxy)

Features X =

1	Recency
2	Frequency
10	TotalQuantity
11	Tenure
12	AvgQuantityPerOrder

Engineered RFM Table

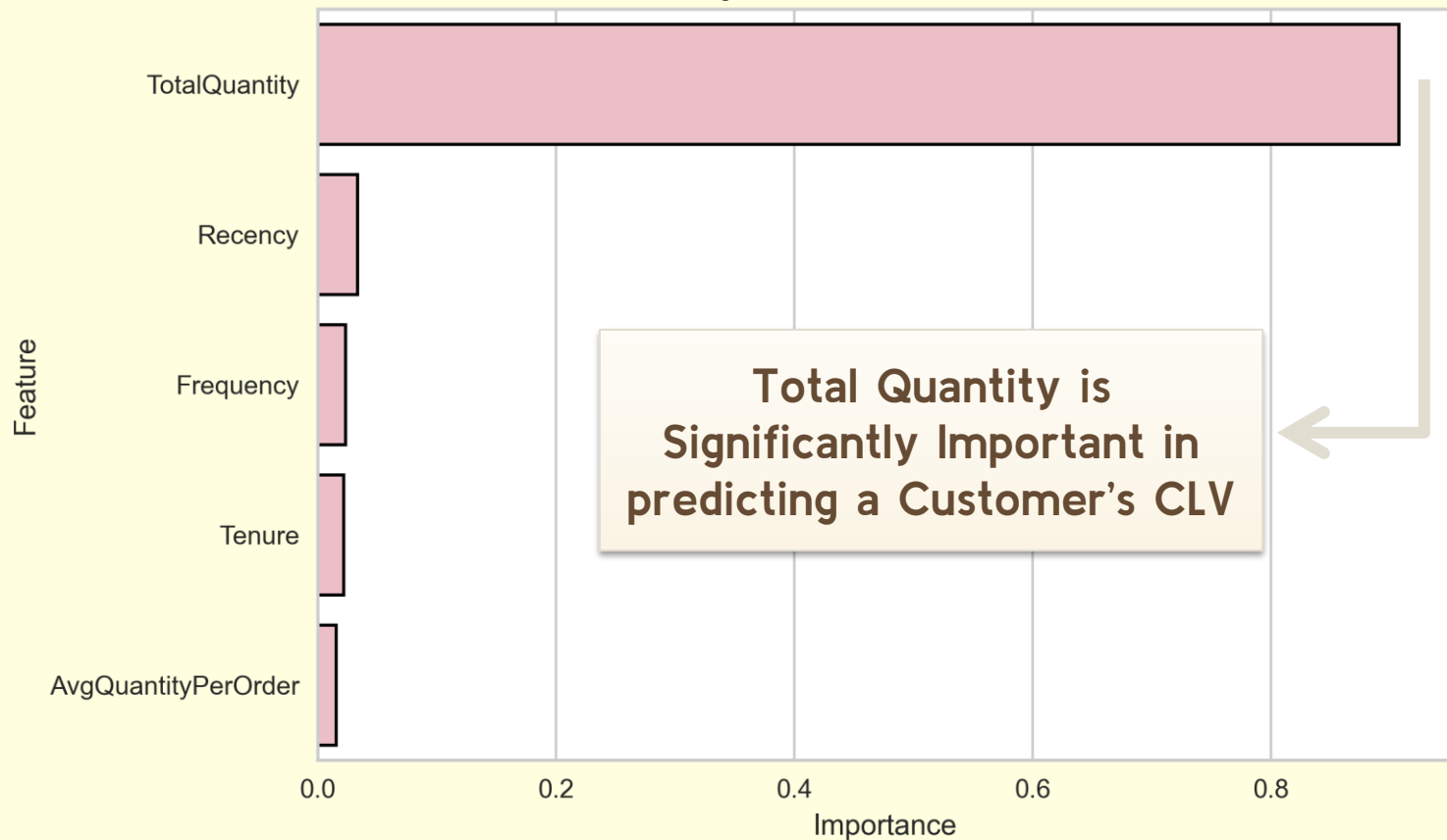
MODEL SELECTION



	Model	RMSE	R2
0	Linear Regression	5713.690743	0.680946
1	Random Forest	5428.893722	0.711960
2	XGBoost	7750.968725	0.412859

- ✓ Lowest Error
- ✓ Highest R² score

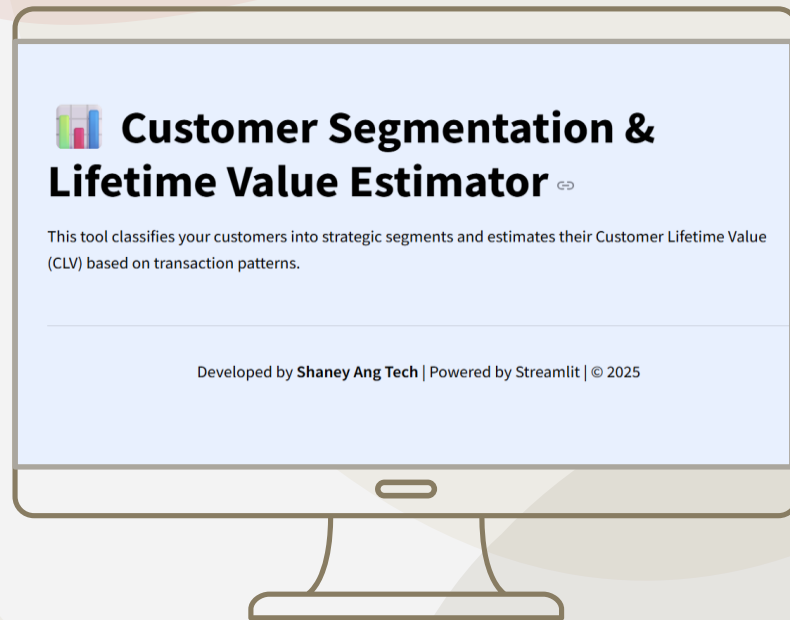
Feature Importance from RandomForest



MODEL DEPLOYMENT DEMO

Streamlit application:

 **Customer Segmentation &
Lifetime Value Estimator**





Customer Purchase Details

Previous Purchase Date

2024/12/24

Most Recent Purchase Date

2025/01/01



Recency (days since last purchase): 8

Purchase Frequency



25



Total Monetary Spend (\$)



500.00



Total Quantity Purchased



30



Customer Tenure (days)



200



Generate Insights

Share



Customer Segmentation & Lifetime Value Estimator

This tool classifies your customers into strategic segments and estimates their Customer Lifetime Value (CLV) based on transaction patterns.



Predicted Segment & Value



Customer Segment: ● VIPs

Top-tier loyal customers. Prioritize retention.



Predicted Lifetime Value (CLV)

\$497.63

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< Manage app



Customer Purchase Details

Previous Purchase Date

2024/07/11

Most Recent Purchase Date

2025/01/01



Recency (days since last purchase):

174

Purchase Frequency



10

-

+

Total Monetary Spend (\$)



200.00

-

+

Total Quantity Purchased



30

-

+

Customer Tenure (days)



200

-

+



Generate Insights

Share



Customer Segmentation & Lifetime Value Estimator

This tool classifies your customers into strategic segments and estimates their Customer Lifetime Value (CLV) based on transaction patterns.



Predicted Segment & Value



Customer Segment: ● At-Risk

These customers may be slipping away. Consider re-engagement campaigns.



Predicted Lifetime Value (CLV)

\$203.88

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< Manage app

INSIGHTS



Key Findings:

1. Our most valuable customers are clustered into the VIPs segment. They spend frequently and in good amount, i.e. high average order value (AOV).
2. RFM features are useful in creating meaningful customer segments and predicting customer values.
3. We can use the predicted CLV, to prioritise retention by relating it to the AOV of each segment.



RECOMMENDATIONS

01

Segment	% of Customers	Avg Order Value	Key Traits
Cluster 3 – VIPs	18%	\$594	Frequent, recent, high spend
Cluster 2 – High Potential	27%	\$456	Moderate recency, good spend but less frequent
Cluster 0 – New/Moderate Buyers	21%	\$222	Recent, low frequency, low-mid spend
Cluster 1 – At-Risk	34%	\$230	Inactive, low frequency, low spend

Strategic Action

Loyalty rewards, VIP perks

Target with special deals or personalized content

Nurture campaigns, upsell

Win-back or exclude from targeting

02 Utilise the Customer Segmentation & CLV Estimator to enhance / prioritise your marketing strategies.

THANK YOU!

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